

SECSR2043 OPERATING SYSTEMS
[20 Marks]

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 Section : 02

Marks

Instruction: Please answer all of the following questions. Whenever the 🙋 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: *ahmad.sh*).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*. *.; cp ../fruits/*. *.;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]



Print screen the script that you type;

```

kahsin@secr2043: ~
GNU nano 7.2      kahsin.sh *
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.c .; cp ../fruits/*.c .;
cd ../*.jar .|

```

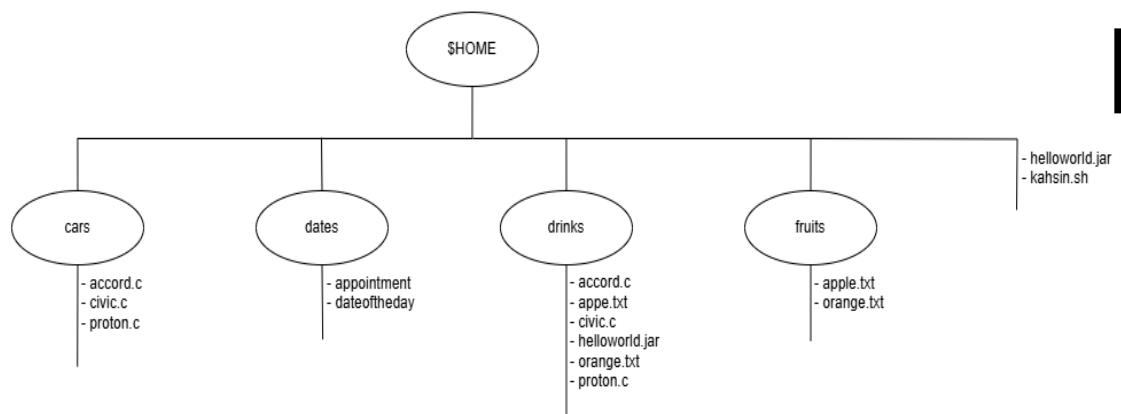
Then draw the tree

```

kahsin@secr2043:~$ tree -I 'snap'
.
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
|-- helloworld.jar
`-- kahsin.sh

5 directories, 15 files

```



- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter “c” as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```

GNU nano 7.2 filecounter.sh
#!/bin/bash

# Ask user for file extension
read -p "Enter the file extension (without dot): " ext

echo "Listing all .$ext files in current directory and subdirectories:"
echo "-----"

# Find and list the files
find . -type f -name ".*$ext"

# Count the files
count=$(find . -type f -name ".*$ext" | wc -l)

echo "-----"
echo "Total number of .$ext files found: $count"

```

```

kahsin@secur2043:~$ nano filecounter.sh
kahsin@secur2043:~$ chmod +x filecounter.sh
kahsin@secur2043:~$ ./filecounter.sh
Enter the file extension (without dot): c
Listing all .c files in current directory and subdirectories:
-----
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
-----
Total number of .c files found: 6

```

2. The following Figure 1 illustrates a tree structure of some directories and files.

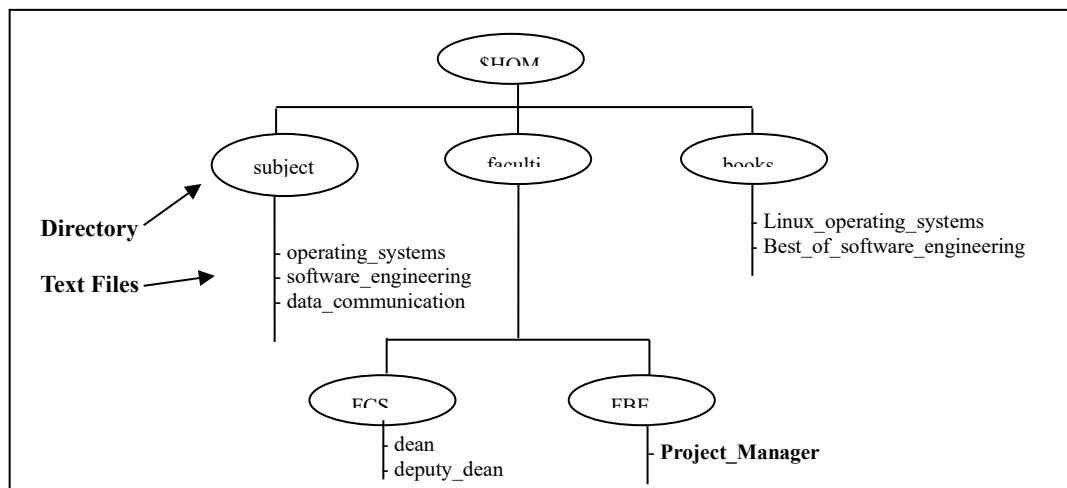


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run

```

GNU nano 7.2                                kahsin2a.sh *
#!/bin/bash

# create top-level directories
mkdir -p $HOME/subject
mkdir -p $HOME/faculti/FCS
mkdir -p $HOME/faculti/FBE
mkdir -p $HOME/books

# create text files inside subject
echo "operating systems" > $HOME/subject/operating_systems
echo "software engineering" > $HOME/subject/software_engineering
echo "data communication" > $HOME/subject/data_communication

# create text files inside faculty/ECS
echo "dean" > $HOME/faculti/FCS/dean
echo "deputy dean" > $HOME/faculti/FCS/deputy_dean

# create text file inside faculty/FRF
echo "Project Manager" > $HOME/faculti/FBE/Project_Manager

# create text files inside books
echo "Linux operating systems" > $HOME/books/Linux_operating_systems
echo "Best of software engineering" > $HOME/books/Best_of_software_engineering

```

```

kahsin@secur2043:~$ nano kahsin2a.sh
kahsin@secur2043:~$ chmod +x kahsin2a.sh
kahsin@secur2043:~$ ./kahsin2a.sh
kahsin@secur2043:~$ tree $HOME -I 'snap'
/home/kahsin
|-- books
|   |-- Best_of_software_engineering
|   |-- Linux_operating_systems
|-- cars
|   |-- accord.c
|   |-- civic.c
|   |-- proton.c
|-- dates
|   |-- appointment
|   |-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   |-- proton.c
|-- faculti
|   |-- FBE
|   |   |-- Project_Manager
|   |-- FCS
|   |   |-- dean
|   |   |-- deputy_dean
|-- filecounter.sh
|-- fruits
|   |-- apple.txt
|   |-- orange.txt
|-- helloworld.jar
|-- kahsin.sh
|-- kahsin2a.sh
|-- kahsin2c.sh
|-- subject
|   |-- data_communication
|   |-- operating_systems
|   |-- software_engineering

```

10 directories, 26 files



- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called `book`.

[2 marks]

Directory/File	Access Control
books	<code>drwxr-xr-x</code>
subject	<code>drwxr-xr-x</code>
Best_of_software_engineering	<code>-rw-r--r--</code>
FCS	<code>drwxr-xr-x</code>
project_manager	<code>-rw-r--r--</code>



```

kahsin@secr2043:~$ ls -ld $HOME/books
ls -ld $HOME/subject
ls -l $HOME/books/Best_of_software_engineering
ls -ld $HOME/faculti/FCS
ls -l $HOME/faculti/FRF/Project_Manager
drwxr-xr-x 2 kahsin kahsin 4096 Jun 29 09:55 /home/kahsin/books
drwxr-xr-x 2 kahsin kahsin 4096 Jun 29 09:55 /home/kahsin/subject
-rw-r--r-- 1 kahsin kahsin 29 Jun 29 10:08 /home/kahsin/books/Best_of_software_engineering
drwxr-xr-x 2 kahsin kahsin 4096 Jun 29 10:07 /home/kahsin/faculti/FCS
-rw-r--r-- 1 kahsin kahsin 16 Jun 29 10:08 /home/kahsin/faculti/FRF/Project_Manager

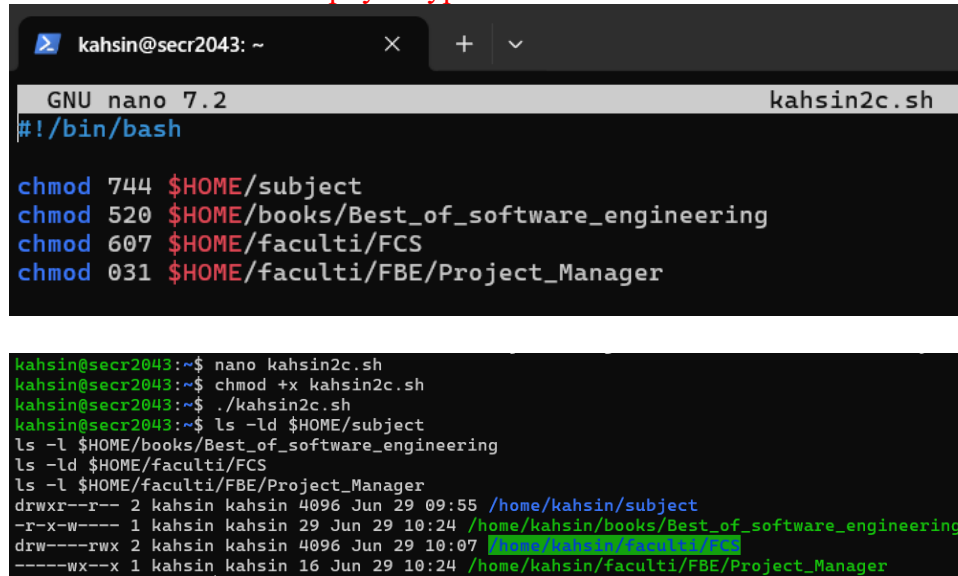
```

- c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subject	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run



```

khsin@seer2043: ~
GNU nano 7.2                                kahsin2c.sh
#!/bin/bash

chmod 744 $HOME/subject
chmod 520 $HOME/books/Best_of_software_engineering
chmod 607 $HOME/faculti/FCS
chmod 031 $HOME/faculti/FBE/Project_Manager

khsin@seer2043:~$ nano kahsin2c.sh
khsin@seer2043:~$ chmod +x kahsin2c.sh
khsin@seer2043:~$ ./kahsin2c.sh
khsin@seer2043:~$ ls -ld $HOME/subject
ls -ld $HOME/books/Best_of_software_engineering
ls -ld $HOME/faculti/FCS
ls -ld $HOME/faculti/FBE/Project_Manager
drwxr--r-- 2 khsin khsin 4096 Jun 29 09:55 /home/khsin/subject
-r-x-w---- 1 khsin khsin 29 Jun 29 10:24 /home/khsin/books/Best_of_software_engineering
drw----rwx 2 khsin khsin 4096 Jun 29 10:07 /home/khsin/faculti/FCS
-----wx--x 1 khsin khsin 16 Jun 29 10:24 /home/khsin/faculti/FBE/Project_Manager

```



- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subject	drwxr--r--
Best_of_software_engineering	-r-x-w----
FCS	drw----rwx
project_manager	-----wx--x

End of Lab 3

*** All the Best for Final Exam ***